

Statistical Methods II

Final Project - Multiple Linear Regression

Step 1: Find Data Summary

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Dataset Name:

Used Cars – Uncovering Factors That Affect Used Car Prices

Dataset Source:

Main Source: <https://data.world/data-society/used-cars-data>

Dataset Authors:

@Data-Society (Original dataset compiled from eBay Kleinanzeigen listings)

Description:

This dataset contains information on used car listings in Germany collected over several years using web scraping techniques. The dataset includes variables such as vehicle price, year of registration, mileage (in kilometers), and engine power (measured in metric horsepower, PS). These variables provide insight into real-world vehicle characteristics and allow for analysis of factors that influence used car prices.

Although the dataset originates from Germany, most variables are numerical and universally interpretable. Some categorical variables require minor translation (e.g., gearbox and seller type), but this does not affect the overall analysis.

Data Cleaning and Preparation

As mentioned previously, the original data set contained many variables, many of which were not relevant to the analysis. An initial data wrangling step was performed to remove unnecessary columns such as advertisement metadata and non-informative attributes.

Column names were standardized for readability, and key variables were renamed (e.g. yearOfRegistration to model_year, powerPS to ps, and kilometer to km).

Invalid and unrealistic values were removed to improve data quality. Specifically, observations with extremely low or high prices, unrealistic engine power values, and outlier registration years were filtered out.

Categorical variables such as gearbox and seller were translated from German to English and standardized into consistent categories. Missing values and empty entries were handled by converting them to NA and removing incomplete observations.

Additional variables derived, including horsepower, mileage in miles, and price in USD, were created for interpretability; however, the analysis will be conducted using the original variables to maintain consistency.

The final dataset after cleaning had **~246,723** observations in total which complies with the $n \geq 25$ needed.

Response Variable (Y):

- price (euros) - *The listed selling price of the vehicle*

Explanatory Variables (X):

Primary (Quantitative):

- yearOfRegistration - *The year the vehicle was first registered*
- kilometer - *The total distance driven by the vehicle (in kilometers)*
- powerPS - *Engine power measured in metric horsepower (PS)*

Additional (Categorical, binary):

- gearbox - *Transmission type (manual or automatic)*
- **seller** - *Seller type (private or commercial, 99.9% of dataset is private sellers so this variable is becoming redundant)*

Additional Variables Considered:

The dataset contains several additional categorical variables such as vehicleType, fuelType, brand, and notRepairedDamage. While these variables may provide further insight into vehicle pricing, some contain multiple categories or require additional preprocessing.

These variables may be explored in other models I have run to evaluate whether they improve model performance (e.g. R-squared) but are not included in the primary model for simplicity and greater interpretability.

Above all, after data cleaning I was still left with units of measurement I couldn't quite grasp as well (i.e. euros, kilometers, Pferdestärke (PS)) so I decided to make some derived variables such as price_usd, miles, and horsepower. These units are not used in our model however to help visualize the data better for those that use the **customary unit's system** as we do in the United States.

Purpose of the Study:

The purpose of this study is to determine which factors significantly affect the price of used cars. Specifically, the goal is to evaluate how variables such as vehicle age, mileage, engine power, transmission type, and seller type contribute to variations in price. Using a multiple linear regression model, this analysis will assess how well these variables can explain and predict used car prices. My analysis will include at least 2 or 3 different MLR models to compare each model's performance based on different variables used.

Reason for Selection:

This dataset was selected due to its real-world relevance and the availability of multiple meaningful variables suitable for regression analysis. Additionally, personal interest in the automotive market and recent fluctuations in used car prices motivated the selection of this topic. The dataset provides an opportunity to explore how vehicle characteristics influence pricing trends in a realistic market setting.